The Pirate Cove Puzzle

Whodunnit? · Basics & Reasoning · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Basics & Reasoning	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Segment addition, midpoint and distance, angle relationships and counting segments.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Navigator Dax took it, in the Treasure Cave, using this method: Sealed in a message bottle.

The reveal (read aloud): While the crew was busy, **Navigator Dax** sneaked into the **Treasure Cave**, sealed the Golden Compass in a **message bottle**, and tossed it into the waves toward his own pirate friends. Every alibi and method you cleared led to him. Yo-ho-ho, great work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: First Mate Morgan · Place: Crow’s Nest
2	Method: Buried in the sand
3	Suspect: Salty Sue · Place: Galley
4	Method: Hidden in an apple barrel
5	Suspect: Polly Finch · Place: Cargo Hold
6	Method: Strapped to the parrot
7	Suspect: Deckhand Finn · Place: Plank Deck
8	Method: Wrapped in a sail
9	Suspect: Cora Blake · Place: Captain’s Cabin
10	Method: Rowed off in a dinghy

Answer Key

The Pirate Cove Puzzle · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	M is the midpoint of AB. AM = 3x + 2 and MB = 5x − 6. Find the length of AB.	**B.** 28	3x + 2 = 5x − 6, so x = 4 AM = 14, so AB = 2(14) 28	First Mate Morgan was scanning the horizon from the Crow's Nest. Two sailors waved to him.
2	Two angles are complementary. One measures x° and the other measures (2x + 15)°. Find the larger angle.	**A.** 65°	x + 2x + 15 = 90, so x = 25 larger = 2(25) + 15 65°	The beach was combed with a metal detector. Nothing buried in the sand.
3	Two angles are supplementary and measure (3x + 10)° and (5x − 6)°. Find the smaller angle.	**D.** 76°	8x + 4 = 180, so x = 22 angles: 76° and 104° 76°	Salty Sue was stirring stew in the Galley. Half the crew came by for seconds.
4	Ray BD bisects ∠ABC. m∠ABD = (3x + 5)° and m∠DBC = (5x − 19)°. Find m∠ABC.	**C.** 82°	3x + 5 = 5x − 19, so x = 12 ∠ABD = 41°, so ∠ABC = 2(41) 82°	The apple barrel was emptied: only apples and one very surprised mouse.
5	Find the distance between (1, 2) and (4, 6).	**C.** 5	d = √(3² + 4²) = √25 5	Polly Finch was feeding the parrot in the Cargo Hold with a crate of crackers.
6	Find the midpoint of the segment with endpoints (−2, 6) and (8, −2).	**A.** (3, 2)	x: $\frac{-2 + 8}{2}$ = 3, y: $\frac{6 + (-2)}{2}$ = 2 (3, 2)	The parrot can only carry crackers. No compass strapped to it.
7	Two vertical angles measure (4x − 7)° and (3x + 14)°. Find the measure of each angle.	**A.** 77°	4x − 7 = 3x + 14, so x = 21 4(21) − 7 77°	Deckhand Finn was scrubbing the Plank Deck. The soap-bubble trail proves it.
8	Points A, B, C are collinear with B between A and C. AB = 12, BC = 2x + 1 and AC = 5x − 2. Find BC.	**D.** 11	12 + 2x + 1 = 5x − 2, so x = 5 BC = 2(5) + 1 11	The sails were unrolled and shaken out. Nothing was wrapped inside.
9	Two angles form a linear pair and measure (2x + 20)° and (4x − 2)°. Find the larger angle.	**B.** 106°	6x + 18 = 180, so x = 27 angles: 74° and 106° 106°	Cora Blake was reading maps with the Captain in the Captain's Cabin.
10	How many different line segments can be named using any two of 6 collinear points as endpoints?	**C.** 15	each pair of points makes one segment: $\frac{6 \cdot 5}{2}$ 15	The dinghy was tied up and dry all night. No one rowed away.

Wrong-Answer Map

The Pirate Cove Puzzle · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. 52	Solved 2x=8 without halving, so x=8	Every message bottle on the ship was held up to the light. None had a compass inside.
1	C. 4	Gave x instead of AB	First Mate Morgan was mapping the tide pools in the Treasure Cave. His boots are still soggy.
1	D. 14	Found AM only, forgot to double	Navigator Dax was checking the stars from the Crow’s Nest. The lookout swears he never climbed down.
2	B. 85°	Sign error in solving for x	The message bottles on the shelf are all corked with dusty wax. None was opened or resealed.
2	C. 125°	Used supplementary (sum 180) instead of complementary	Salty Sue was hunting for crab shells in the Treasure Cave. Her full basket proves it.
2	D. 25°	Gave the smaller angle	Navigator Dax was polishing the Captain’s telescope in the Captain’s Cabin all evening.
3	A. 22°	Gave x instead of the angle	Polly Finch was teaching the parrot to squawk in the Treasure Cave. The echo was deafening.
3	B. 79°	Arithmetic slip: 8x−4=180	The bottle washed up on shore holds only a note about lunch. No compass was sealed inside.
3	C. 104°	Gave the larger angle	Navigator Dax was tasting soup in the Galley. The cook gave him two bowls.
4	A. 12°	Gave x instead of the angle	Deckhand Finn was lighting lanterns in the Treasure Cave. The wax drips are still warm.
4	B. 164°	Doubled the whole angle twice	The neck of the message bottle is too narrow for the Golden Compass to fit through.
4	D. 41°	Gave m∠ABD only, forgot to double	Navigator Dax was counting barrels in the Cargo Hold with the quartermaster.
5	A. 7	Added the legs instead of using the distance formula	Navigator Dax was swabbing the Plank Deck beside Deckhand Finn. The mop buckets prove it.
5	B. 10	Added the squared differences wrongly (doubled)	The harbor master collected every floating bottle at dawn. All were empty.
5	D. 25	Forgot the square root	Cora Blake was sketching stalactites in the Treasure Cave. Her charcoal smudges are everywhere.
6	B. (6, 4)	Added coordinates without dividing by 2	Navigator Dax was waving flags from the Crow’s Nest. A passing ship waved back.
6	C. (5, −4)	Halved the differences instead of the sums	First Mate Morgan was chasing a crab around the Treasure Cave. The crab won.
6	D. (3, 4)	Dropped the negative when adding y-values	The ink on the bottle’s note is dry and old. No one sealed anything in it last night.
7	B. 154°	Added the two angles	Salty Sue was gathering seaweed in the Treasure Cave. Her bucket was full when she came out.
7	C. 21°	Gave x instead of the angle	Navigator Dax was signing logbooks in the Captain’s Cabin. The Captain stamped each page.
7	D. 103°	Treated vertical angles as supplementary	The cork in every message bottle is cracked and loose. None could stay sealed at sea.
8	A. 23	Gave AC instead of BC	Polly Finch was whistling sea shanties in the Treasure Cave. The echoes gave her away.

8	B. 13	Added 12 + 1 only	The message bottles are tied in a net below deck. The knots were never touched.
8	C. 5	Gave x instead of BC	Navigator Dax was peeling potatoes in the Galley. The pile grew taller by the minute.
9	A. 74°	Gave the smaller angle	Navigator Dax was stacking crates in the Cargo Hold. The crew counted every one.
9	C. 46°	Used 90 instead of 180 for a linear pair	The sea was too rough to float any bottle last night. Not one left the ship.
9	D. 27°	Gave x instead of the angle	Deckhand Finn was digging for shells in the Treasure Cave. Sand still fills his boots.
10	A. 30	Counted ordered pairs (AB and BA)	Navigator Dax was fishing off the Plank Deck. The ship’s cat watched him all night.
10	B. 36	Squared the number of points	The last message bottle was found full of seawater. Nothing dry could have been sealed in it.
10	D. 21	Added 1 too many: 6+5+4+3+2+1	Cora Blake was resting on a rock in the Treasure Cave, waiting for the tide to turn.